week6_notebook_student

November 11, 2026

1 HS4002 Week 6

1.1 OLS with Higher-Order Effects, Interaction Terms, and Logistic Regression

Today we cover: 1. Polynomial terms in OLS (age-squared) 2. Interaction effects 3. The Linear Probability Model (LPM) 4. Logistic regression 5. Marginal effects

[24]:
```
%pip install pandas numpy plotnine statsmodels marginaleffects stargazer

import pandas as pd
import numpy as np
from plotnine import *
import warnings
warnings.filterwarnings('ignore')
import statsmodels.formula.api as smf
from statsmodels.stats.anova import anova_lm
```

```
Requirement already satisfied: pandas in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (3.0.5)
Requirement already satisfied: numpy in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (2.5.3)
Requirement already satisfied: plotnine in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (0.15.8)
Requirement already satisfied: statsmodels in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (0.15.0)
Requirement already satisfied: marginaleffects in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (0.6.1)
Requirement already satisfied: stargazer in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (0.0.7)
Requirement already satisfied: python-dateutil>=2.8.2 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from pandas) (2.9.0.post0)
Requirement already satisfied: matplotlib>=3.8.0 in
```

```
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from plotnine) (3.11.2)
Requirement already satisfied: mizani~=0.14.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from plotnine) (0.14.6)
Requirement already satisfied: scipy>=1.8.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from plotnine) (1.18.1)
Requirement already satisfied: patsy>=0.5.6 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from statsmodels) (1.0.3)
Requirement already satisfied: packaging>=21.3 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from statsmodels) (26.3)
Requirement already satisfied: formulaic>=1.1.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from statsmodels) (1.2.2)
Requirement already satisfied: jax>=0.4.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (0.11.1)
Requirement already satisfied: narwhals>=1.34.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (2.26.0)
Requirement already satisfied: polars>=1.26.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (1.44.2)
Requirement already satisfied: pydantic>=2.10.3 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (2.13.5)
Requirement already satisfied: pyarrow>=19.0.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (25.0.1)
Requirement already satisfied: interface-meta>=1.2.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from formulaic>=1.1.0->statsmodels) (2.0.1)
Requirement already satisfied: typing-extensions>=4.2.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from formulaic>=1.1.0->statsmodels) (4.16.0)
Requirement already satisfied: wrapt>=1.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from formulaic>=1.1.0->statsmodels) (2.4.1)
Requirement already satisfied: jaxlib<=0.11.1,>=0.11.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from jax>=0.4.0->marginaleffects) (0.11.1)
Requirement already satisfied: ml_dtypes>=0.5.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from jax>=0.4.0->marginaleffects) (0.6.0)
Requirement already satisfied: opt_einsum in
```

```
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from jax>=0.4.0->marginaleffects) (3.4.0)
Requirement already satisfied: contourpy>=1.0.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine) (1.4.0)
Requirement already satisfied: cycler>=0.10 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine) (4.65.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine) (1.5.1)
Requirement already satisfied: pillow>=9 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine) (12.3.0)
Requirement already satisfied: pyparsing>=3 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine) (3.3.2)
Requirement already satisfied: polars-runtime-32==1.44.2 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from polars>=1.26.0->marginaleffects) (1.44.2)
Requirement already satisfied: annotated-types>=0.6.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from pydantic>=2.10.3->marginaleffects) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.5 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from pydantic>=2.10.3->marginaleffects) (2.46.5)
Requirement already satisfied: typing-inspection>=0.4.2 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from pydantic>=2.10.3->marginaleffects) (0.4.4)
Requirement already satisfied: six>=1.5 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from python-dateutil>=2.8.2->pandas) (1.17.0)

[notice] A new release of pip is
available: 24.2 -> 26.3
[notice] To update, run:
pip install --upgrade pip
Note: you may need to restart the kernel to use updated packages.
```

[25]:
```
raw_df = pd.read_csv("gss2022_mini.csv")

chosen_variables = [
        'id', 'age', 'yrlvmus', 'yrartxbt', 'yrmovie', 'yrcreat',
        'income16', 'prestg10', 'degree', 'race', 'sex', 'wrkstat'
```

```
]
df = raw_df[chosen_variables].dropna().copy()

# Recode income16 to continuous midpoint values
income_map = {
        1: 500,    2: 2000,   3: 3500,   4: 4500,   5: 5500,   6: 6500,
        7: 7500,   8: 9000,   9: 11250,  10: 13750, 11: 16250, 12: 18750,
        13: 21250, 14: 23750, 15: 27500, 16: 32500, 17: 37500, 18: 45000,
        19: 55000, 20: 67500, 21: 82500, 22: 100000, 23: 120000, 24: 140000,
        25: 160000, 26: 250000
}
df['income_cont'] = df['income16'].map(income_map)

# Binary cultural participation indicators
df['binary_lvmus']  = (df['yrlvmus']  == 1).astype(int)
df['binary_artxbt'] = (df['yrartxbt'] == 1).astype(int)
df['binary_movie']  = (df['yrmovie']  == 1).astype(int)
df['binary_creat']  = (df['yrcreat']  == 1).astype(int)
df['omni'] = df[['binary_lvmus','binary_artxbt','binary_movie','binary_creat']].
 ↪sum(axis=1)

# Derived variables
df['race_bin']   = df['race'].astype(str)
df['ba_binary']  = (df['degree'] >= 3).astype(int)
df['sex_woman']  = np.where(df['sex'] == 2, 'Woman', 'Not Woman')
df['working']    = np.where(df['wrkstat'] <= 3, 'Working', 'Not Working')
```

2 OLS with Higher-Order Polynomials

Begin by regressing income on age. Save as `ols_model1`.

[26]:
```
ols_model1 = smf.ols('income_cont ~ age', data=df).fit()
ols_model1.summary()
# interpretation: We observe no associations between age and income (p = 0.892)␣
 ↪, ceteris paribus
```

[26]:

Dep. Variable:	income_cont	**R-squared:**	0.000
Model:	OLS	**Adj. R-squared:**	-0.001
Method:	Least Squares	**F-statistic:**	0.01847
Date:	Wed, 11 Nov 2026	**Prob (F-statistic):**	0.892
Time:	23:10:38	**Log-Likelihood:**	-8910.5
No. Observations:	707	**AIC:**	1.782e+04
Df Residuals:	705	**BIC:**	1.783e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	8.632e+04	8546.726	10.099	0.000	6.95e+04	1.03e+05
age	21.2431	156.330	0.136	0.892	-285.684	328.171

Omnibus:	113.766	**Durbin-Watson:**	1.629
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	168.872
Skew:	1.178	**Prob(JB):**	2.14e-37
Kurtosis:	3.430	**Cond. No.**	172.

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

2.1 Adding an age-squared term

In Python formulas, use `I(age**2)` — the `I()` wrapper tells the formula parser to treat the expression literally.

[27]:
```
# this is assuming it is not a linear model! the age^2 allows line to bend
# adding age^2 asks if income goes up with age, then comes down
ols_model2 = smf.ols('income_cont ~ age + I(age**2)', data=df).fit()
ols_model2.summary()
# income = aage^2 + bage + c
# income = -37.5age^2 + 3886age - 2088
# peak age occurs when rate of change is 0. dy/dx = 0
# dy/dx = 2(-37.5)age + 3886 = 0
# peak_age = -3886/2(-37.5) = 52 (round up)
# Income rises with age until about 52, then gently falls. The pattern is␣
 ↪statistically real, but age alone explains only about 2% of why people's␣
 ↪incomes differ.
```

[27]:

Dep. Variable:	income_cont	**R-squared:**	0.021
Model:	OLS	**Adj. R-squared:**	0.018
Method:	Least Squares	**F-statistic:**	7.568
Date:	Wed, 11 Nov 2026	**Prob (F-statistic):**	0.000560
Time:	23:10:38	**Log-Likelihood:**	-8903.0
No. Observations:	707	**AIC:**	1.781e+04
Df Residuals:	704	**BIC:**	1.783e+04
Df Model:	2		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-2088.6030	2.43e+04	-0.086	0.931	-4.97e+04	4.55e+04
age	3886.3291	1006.049	3.863	0.000	1911.113	5861.545
I(age ** 2)	-37.4677	9.636	-3.888	0.000	-56.387	-18.548

Omnibus:	109.972	**Durbin-Watson:**	1.611
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	160.918
Skew:	1.148	**Prob(JB):**	1.14e-35
Kurtosis:	3.440	**Cond. No.**	3.16e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.16e+04. This might indicate that there are strong multicollinearity or other numerical problems.

2.2 Making a regression table

As in Week 5, the `stargazer` package collects fitted models into a single regression table. Reading models side by side is much easier than scrolling through two separate `.summary()` blocks.

[28]:
```
from stargazer.stargazer import Stargazer

sg = Stargazer([ols_model1, ols_model2])
sg.title('Income, Age, and Age-Squared')

# Label the models and variables readably, rather than by raw column name
sg.dependent_variable_name('Income (USD)')

sg.custom_columns(['Model 1', 'Model 2'], [1, 1])
sg.covariate_order(['age', 'I(age ** 2)', 'Intercept'])
sg.rename_covariates({
        'age':          'Age',
        'I(age ** 2)': 'Age-squared',
        'Intercept':   'Intercept'
})

# Ending the cell on the object itself renders the table in the notebook
sg
```

[28]:
```
<stargazer.stargazer.Stargazer at 0x13ece6fc0>
```

[29]:
```
person_a = 15
person_b = 45
(person_b - person_a)*3886 + ((person_b - person_a)**2)*(-37.5)
# 3886 is the main effect, 37.5 is the quadratic effect
# Person B is expected to have $82830 more than Person B
```

[29]:
```
82830.0
```

2.3 Interaction Effects

Add BA education and gender to the model. Save as `ols_model3`.

[30]:
```
ols_model3 = smf.ols('income_cont ~ age + I(age**2) + ba_binary + sex_woman',␣
↪data=df).fit()
ols_model3.summary()
```

[30]:

Dep. Variable:	income_cont	**R-squared:**	0.203
Model:	OLS	**Adj. R-squared:**	0.199
Method:	Least Squares	**F-statistic:**	44.79
Date:	Wed, 11 Nov 2026	**Prob (F-statistic):**	1.61e-33
Time:	23:10:38	**Log-Likelihood:**	-8830.1
No. Observations:	707	**AIC:**	1.767e+04
Df Residuals:	702	**BIC:**	1.769e+04
Df Model:	4		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-8893.2601	2.21e+04	-0.402	0.688	-5.23e+04	3.46e+04
sex_woman[T.Woman]	-1.663e+04	4871.657	-3.414	0.001	-2.62e+04	-7067.441
age	3275.5520	910.156	3.599	0.000	1488.599	5062.505
I(age ** 2)	-31.3456	8.720	-3.595	0.000	-48.466	-14.225
ba_binary	6.04e+04	4874.146	12.393	0.000	5.08e+04	7e+04

Omnibus:	102.337	**Durbin-Watson:**	1.765
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	145.602
Skew:	1.046	**Prob(JB):**	2.42e-32
Kurtosis:	3.751	**Cond. No.**	3.19e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.19e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Now add an interaction term between BA and gender. In statsmodels formulas, `:` creates an interaction.

Save as `ols_model4`.

[31]:
```
# this model asks if education pays off equally for men and women.
ols_model4 = smf.ols(
        'income_cont ~ age + I(age**2) + ba_binary + sex_woman + ba_binary:
 ↪sex_woman',
        data=df
).fit()
ols_model4.summary()
# education pays off less for women. for men, education adds $69623 to their␣
 ↪income
# but for women, education adds $69623-$17474=$52149 to their income.
# but p > 0.1, so evidence is weak
```

[31]:

Dep. Variable:	income_cont	**R-squared:**	0.207
Model:	OLS	**Adj. R-squared:**	0.201
Method:	Least Squares	**F-statistic:**	36.59
Date:	Wed, 11 Nov 2026	**Prob (F-statistic):**	2.37e-33
Time:	23:10:38	**Log-Likelihood:**	-8828.5
No. Observations:	707	**AIC:**	1.767e+04
Df Residuals:	701	**BIC:**	1.770e+04
Df Model:	5		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P> |t|**	**[0.025**	**0.975]**
Intercept	-1.184e+04	2.22e+04	-0.534	0.593	-5.53e+04	3.17e+04
sex_woman[T.Woman]	-8323.3760	6712.340	-1.240	0.215	-2.15e+04	4855.323
age	3221.5326	909.213	3.543	0.000	1436.425	5006.640
I(age ** 2)	-30.7979	8.712	-3.535	0.000	-47.902	-13.694
ba_binary	6.962e+04	7073.827	9.843	0.000	5.57e+04	8.35e+04
ba_binary:sex_woman[T.Woman]	-1.748e+04	9733.988	-1.796	0.073	-3.66e+04	1626.933

Omnibus:	102.870	**Durbin-Watson:**	1.768
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	146.972
Skew:	1.043	**Prob(JB):**	1.22e-32
Kurtosis:	3.796	**Cond. No.**	3.21e+04

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.21e+04. This might indicate that there are
strong multicollinearity or other numerical problems.

[32]:
```
from stargazer.stargazer import Stargazer

sg = Stargazer([ols_model3, ols_model4])
sg.title('Interaction Between BA and Gender')
sg.dependent_variable_name('Income (USD)')

sg.custom_columns(['Model 3', 'Model 4'], [1, 1])
sg.covariate_order([
        'age',
```

```
        'I(age ** 2)',
        'ba_binary',
        'sex_woman[T.Woman]',
        'ba_binary:sex_woman[T.Woman]',
        'Intercept'
])
sg.rename_covariates({
        'age':                            'Age',
        'I(age ** 2)':                    'Age-squared',
        'ba_binary':                      'BA degree',
        'sex_woman[T.Woman]':             'Gender (woman = 1)',
        'ba_binary:sex_woman[T.Woman]': 'BA × Woman',
        'Intercept':                      'Intercept'
})

sg
```

[32]: <stargazer.stargazer.Stargazer at 0x13eaf82c0>

Apart from the main effects of age, education and gender, we also observe an interaction effect between education and gender. For women, each year of education is associated with a decreased income of $17484 when compared to men ($p < 0.1$).

3 Using the Linear Probability Model (LPM)

The LPM uses OLS on a binary dependent variable. It's quick and interpretable, though it can predict probabilities outside [0, 1].

Regress live music attendance (`binary_lvmus`) on BA education and log income.

[33]:
```
lpm_model1 = smf.ols(
        'binary_lvmus ~ ba_binary + np.log(income_cont)',
        data=df
).fit()
print(lpm_model1.summary())

# with lpm, binary_lvmus only takes 2 values, 0 or 1
# this model is describing the the relationship between income and the␣
 ↪probability of live music attendance
# Having a BA is expected to be associated with a 0.128 (p < 0.01) increase in␣
 ↪the probability of attending live music, holding all else constant
# an increase in one unit of income is expected to be asscoiated with a 0.0887␣
 ↪(p < 0.01) increase in the probability of attending live music, holding all␣
 ↪else constant
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:           binary_lvmus   R-squared:                       0.069
```

```
Model:                            OLS   Adj. R-squared:                  0.066
Method:                 Least Squares   F-statistic:                     26.08
Date:                Wed, 11 Nov 2026   Prob (F-statistic):           1.19e-11
Time:                        23:10:39   Log-Likelihood:                -486.80
No. Observations:                 707   AIC:                             979.6
Df Residuals:                     704   BIC:                             993.3
Df Model:                           2
Covariance Type:            nonrobust
================================================================================
=======
                           coef    std err          t      P>|t|      [0.025
0.975]
--------------------------------------------------------------------------------
-------
Intercept               -0.5077      0.205     -2.471      0.014      -0.911
-0.104
ba_binary                0.1280      0.040      3.218      0.001       0.050
0.206
np.log(income_cont)      0.0887      0.019      4.603      0.000       0.051
0.127
==============================================================================
Omnibus:                     3543.878   Durbin-Watson:                   1.904
Prob(Omnibus):                  0.000   Jarque-Bera (JB):               87.899
Skew:                          -0.118   Prob(JB):                     8.18e-20
Kurtosis:                       1.289   Cond. No.                         126.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly
specified.
```

4 Using a Logit Model

Now, let's try logistic regression.

[34]:
```
# logistic regression helps ensure the y-value stays between 0 and 1 by logging␣
 ↪it
# this model is describing the relationship between income and the log odds of␣
 ↪attendance. thus LPM and logit models are not comparable

# careful: how large the coefficient is does not matter. since model is logged,␣
 ↪it is difficult to predict the impact the size of coefficient has on y
# to understand how coefficient impacts y for logit models, look at marginal␣
 ↪effects instead!
logit_model1 = smf.logit(
        'binary_lvmus ~ ba_binary + np.log(income_cont)',
        data=df
```

```
).fit()
print(logit_model1.summary())

# computer took 5 rounds to guenss the answer
# pseudo R-squared = 0.05151. model improves on a model with no predictors (i.e.
  ↪ LL-Null value) by 5%
```

```
Optimization terminated successfully.
         Current function value: 0.655998
         Iterations 5
                           Logit Regression Results
==============================================================================
Dep. Variable:           binary_lvmus   No. Observations:                  707
Model:                          Logit   Df Residuals:                      704
Method:                           MLE   Df Model:                            2
Date:                Wed, 11 Nov 2026   Pseudo R-squ.:                 0.05151
Time:                        23:10:39   Log-Likelihood:                -463.79
converged:                       True   LL-Null:                       -488.98
Covariance Type:            nonrobust   LLR p-value:                 1.151e-11
==============================================================================
=======
                             coef    std err          z      P>|z|      [0.025
0.975]
------------------------------------------------------------------------------
-------
Intercept              -4.4595      0.961     -4.639      0.000      -6.344
-2.575
ba_binary               0.5186      0.169      3.062      0.002       0.187
0.851
np.log(income_cont)     0.3938      0.090      4.393      0.000       0.218
0.569
==============================================================================
=======
```

a one unit increase in education is expected to cause an increase in the log odds of attending live music by 0.5186 ($p < 0.01$), holding all else constant. a one unit increase in income is expected to cause an increase in the log odds of attending live music by 0.3938($p < 0.001$), holding all else constant.

Build a second logit model adding gender, age, and age-squared. Compare both.

[35]:
```
logit_model2 = smf.logit(
        'binary_lvmus ~ ba_binary + np.log(income_cont) + sex_woman + age +␣
  ↪I(age**2)',
        data=df
).fit()
print(logit_model2.summary())
```

```
Optimization terminated successfully.
```

```
         Current function value: 0.645323
         Iterations 5
                           Logit Regression Results
==============================================================================
Dep. Variable:           binary_lvmus   No. Observations:                  707
Model:                          Logit   Df Residuals:                      701
Method:                           MLE   Df Model:                            5
Date:                Wed, 11 Nov 2026   Pseudo R-squ.:                 0.06695
Time:                        23:10:39   Log-Likelihood:                -456.24
converged:                       True   LL-Null:                       -488.98
Covariance Type:            nonrobust   LLR p-value:                 8.952e-13
==============================================================================
=======
                             coef    std err          z      P>|z|      [0.025
0.975]
------------------------------------------------------------------------------
-------
Intercept              -3.6710      1.156     -3.176      0.001      -5.936
-1.405
sex_woman[T.Woman]      0.4385      0.160      2.743      0.006       0.125
0.752
ba_binary               0.4676      0.172      2.713      0.007       0.130
0.805
np.log(income_cont)     0.4501      0.094      4.813      0.000       0.267
0.633
age                    -0.0581      0.030     -1.937      0.053      -0.117
0.001
I(age ** 2)             0.0005      0.000      1.637      0.102   -9.26e-05
0.001
==============================================================================
=======
```

[36]:
```
# compare lpm and logit models using stargazer

from stargazer.stargazer import Stargazer

sg = Stargazer([lpm_model1, logit_model1, logit_model2])
sg.title('Live Music Attendance: LPM vs Logit')
sg.dependent_variable_name('Attends live music')

sg.custom_columns(['LPM', 'Logit 1', 'Logit 2'], [1, 1, 1])
sg.covariate_order([
        'ba_binary',
        'np.log(income_cont)',
        'sex_woman[T.Woman]',
        'age',
        'I(age ** 2)',
```

```
        'Intercept'
])
sg.rename_covariates({
        'ba_binary':            'BA degree',
        'np.log(income_cont)': 'log(Income)',
        'sex_woman[T.Woman]':  'Gender (woman = 1)',
        'age':                  'Age',
        'I(age ** 2)':          'Age-squared',
        'Intercept':            'Intercept'
})

# The LPM coefficients are changes in probability, the logit coefficients are
# log-odds, so compare signs and significance across columns - not magnitudes.
sg

# the tables look very different as the models are measuring different things␣
 ↪-- one is measuring probability, the rest are measuring log odds
```

[36]: <stargazer.stargazer.Stargazer at 0x13ecf0c50>

4.1 Comparing Logit Models

4.1.1 AIC comparison

[37]:
```
print(f'AIC logit_model1: {logit_model1.aic:.2f}')
print(f'AIC logit_model2: {logit_model2.aic:.2f}')
# model with smaller AIC is better
```

```
AIC logit_model1: 933.58
AIC logit_model2: 924.49
```

4.1.2 Likelihood ratio test

[ ]:
```
# use this to compare logit nested models

from scipy.stats import chi2

lr_stat = 2 * (logit_model2.llf - logit_model1.llf)
df_diff = logit_model2.df_model - logit_model1.df_model
p_value = chi2.sf(lr_stat, df_diff)
print(f'LR statistic = {lr_stat:.4f}')
print(f'df = {df_diff:.0f}')
print(f'p-value = {p_value:.4f}')

# null hypothesis is that smaller model is the true model. if p is small,␣
 ↪reject the null hypothesis
# since p-value is < 0.01, we reject the null hypothesis. adding the variables␣
 ↪gender, age and age-squared made the model fit better
```

```
LR statistic = 15.0945
df = 3
p-value = 0.0017
```

5 Marginal Effects

The Python `marginaleffects` package mirrors R's package of the same name.

[44]:
```
%pip install marginaleffects
from marginaleffects import avg_comparisons, comparisons
```

```
Requirement already satisfied: marginaleffects in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (0.6.1)
Requirement already satisfied: formulaic>=1.0.2 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (1.2.2)
Requirement already satisfied: jax>=0.4.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (0.11.1)
Requirement already satisfied: narwhals>=1.34.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (2.26.0)
Requirement already satisfied: numpy>=2.0.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (2.5.3)
Requirement already satisfied: patsy>=1.0.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (1.0.3)
Requirement already satisfied: polars>=1.26.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (1.44.2)
Requirement already satisfied: pydantic>=2.10.3 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (2.13.5)
Requirement already satisfied: plotnine>=0.14.5 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (0.15.8)
Requirement already satisfied: scipy>=1.14.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (1.18.1)
Requirement already satisfied: pyarrow>=19.0.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from marginaleffects) (25.0.1)
Requirement already satisfied: interface-meta>=1.2.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from formulaic>=1.0.2->marginaleffects) (2.0.1)
Requirement already satisfied: pandas>=1.3 in
```

```
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from formulaic>=1.0.2->marginaleffects) (3.0.5)
Requirement already satisfied: typing-extensions>=4.2.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from formulaic>=1.0.2->marginaleffects) (4.16.0)
Requirement already satisfied: wrapt>=1.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from formulaic>=1.0.2->marginaleffects) (2.4.1)
Requirement already satisfied: jaxlib<=0.11.1,>=0.11.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from jax>=0.4.0->marginaleffects) (0.11.1)
Requirement already satisfied: ml_dtypes>=0.5.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from jax>=0.4.0->marginaleffects) (0.6.0)
Requirement already satisfied: opt_einsum in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from jax>=0.4.0->marginaleffects) (3.4.0)
Requirement already satisfied: packaging in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from patsy>=1.0.1->marginaleffects) (26.3)
Requirement already satisfied: matplotlib>=3.8.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from plotnine>=0.14.5->marginaleffects) (3.11.2)
Requirement already satisfied: mizani~=0.14.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from plotnine>=0.14.5->marginaleffects) (0.14.6)
Requirement already satisfied: statsmodels>=0.14.6 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from plotnine>=0.14.5->marginaleffects) (0.15.0)
Requirement already satisfied: polars-runtime-32==1.44.2 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from polars>=1.26.0->marginaleffects) (1.44.2)
Requirement already satisfied: annotated-types>=0.6.0 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from pydantic>=2.10.3->marginaleffects) (0.8.0)
Requirement already satisfied: pydantic-core==2.46.5 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from pydantic>=2.10.3->marginaleffects) (2.46.5)
Requirement already satisfied: typing-inspection>=0.4.2 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from pydantic>=2.10.3->marginaleffects) (0.4.4)
Requirement already satisfied: contourpy>=1.0.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine>=0.14.5->marginaleffects) (1.4.0)
Requirement already satisfied: cycler>=0.10 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine>=0.14.5->marginaleffects) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in
```

```
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine>=0.14.5->marginaleffects) (4.65.0)
Requirement already satisfied: kiwisolver>=1.3.1 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine>=0.14.5->marginaleffects) (1.5.1)
Requirement already satisfied: pillow>=9 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine>=0.14.5->marginaleffects) (12.3.0)
Requirement already satisfied: pyparsing>=3 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine>=0.14.5->marginaleffects) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from matplotlib>=3.8.0->plotnine>=0.14.5->marginaleffects)
(2.9.0.post0)
Requirement already satisfied: six>=1.5 in
/Users/yu/Documents/hs4002_ay26/hs4002_labs2026/.venv/lib/python3.12/site-
packages (from python-
dateutil>=2.7->matplotlib>=3.8.0->plotnine>=0.14.5->marginaleffects) (1.17.0)

[notice] A new release of pip is
available: 24.2 -> 26.3
[notice] To update, run:
pip install --upgrade pip
Note: you may need to restart the kernel to use updated packages.
```

5.1 Average marginal effect

The average marginal effect of a BA degree — equivalent to R's `avg_comparisons(logit_model1, variables='ba_binary')`.

[ ]:
```
# prefer using this as it makes more sense than imagining a hypothetical guy␣
↪(which is what marginal effect at the mean does)
avg_comparisons(logit_model1, variables='ba_binary')
# average marginal effect for evrey bachelor's degree and get the average of␣
↪that
# contrast: comparing having education (1) with no education (0)
# estimate is the B
# statistic is the z-value
```

[ ]:
```
shape: (1, 7)

 Estimate  Std.Error  z     P(>|z|)  S     2.5%    97.5%
 ---       ---        ---   ---      ---   ---     ---
 str       str        str   str      str   str     str

 0.124     0.0408     3.04  0.00247  8.66  0.0438  0.204
```

```
Term: ba_binary
Contrast: 1 - 0
```

Interpretation: having a BA degree is associated, on average across the sample, with an increase of 0.1238 in the probability of attending live music, holding all else constant. at p = 0.002365, we reject the null hypothesis that there is no effect at the 0.05 threshold.

5.2 Marginal effect at the mean

The marginal effect for a hypothetical person at the average of all X variables.

[41]:
```
comparisons(logit_model1, variables='ba_binary', newdata='mean')
```

[41]:
```
shape: (1, 7)

 Estimate  Std.Error  z     P(>|z|)  S     2.5%    97.5%
 ---       ---        ---   ---      ---   ---     ---
 str       str        str   str      str   str     str

 0.126     0.041      3.08  0.00213  8.87  0.0459  0.207

Term: ba_binary
Contrast: 1 - 0
```

having a BA degree is associated, for a typical average person, with an increase of 0.1264 in the probability of attending live music, holding all else constant. at p = 0.002133, we reject the null hypothesis that there is no effect at the 0.05 threshold

[42]:
```
# average marginal effect of income from logit 1 model
avg_comparisons(logit_model1, variables='income_cont')
```

[42]:
```
shape: (1, 7)

 Estimate  Std.Error  z     P(>|z|)   S     2.5%      97.5%
 ---       ---        ---   ---       ---   ---       ---
 str       str        str   str       str   str       str

 3.25e-06  4.13e-07   7.88  1.24e-14  46.2  2.44e-06  4.06e-06

Term: income_cont
Contrast: +1
```

every unit of income is associated with, on average across the sample, with an increase of 0.000003 in the probability of attending live music, holding all else constant. at p = 1.2434e, we reject the null hypothesis that there is no effect at the 0.05 threshold

[43]:
```
# regressing the impact of sex and income on whether or not you watch a movie

lpm_model5 = smf.ols(
```

```
        'binary_movie ~ sex_woman + np.log(income_cont)',
        data=df
).fit()
print(lpm_model5.summary())
```

```
                            OLS Regression Results
==============================================================================
Dep. Variable:           binary_movie   R-squared:                       0.037
Model:                            OLS   Adj. R-squared:                  0.034
Method:                 Least Squares   F-statistic:                     13.57
Date:                Wed, 11 Nov 2026   Prob (F-statistic):           1.65e-06
Time:                        23:10:42   Log-Likelihood:                -499.73
No. Observations:                 707   AIC:                             1005.
Df Residuals:                     704   BIC:                             1019.
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept     -0.4055      0.201     -2.021      0.044      -0.800      -0.012
sex_woman[T.Woman]      0.1236      0.037      3.331      0.001       0.051       0.196
np.log(income_cont)     0.0761      0.018      4.233      0.000       0.041       0.111
==============================================================================
Omnibus:                     3073.752   Durbin-Watson:                   1.925
Prob(Omnibus):                  0.000   Jarque-Bera (JB):              101.124
Skew:                           0.019   Prob(JB):                     1.10e-22
Kurtosis:                       1.148   Cond. No.                         121.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```